

Lab 1: Zabbix Installation

Durham College

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INFT1202: Network Monitoring and Penetration Testing

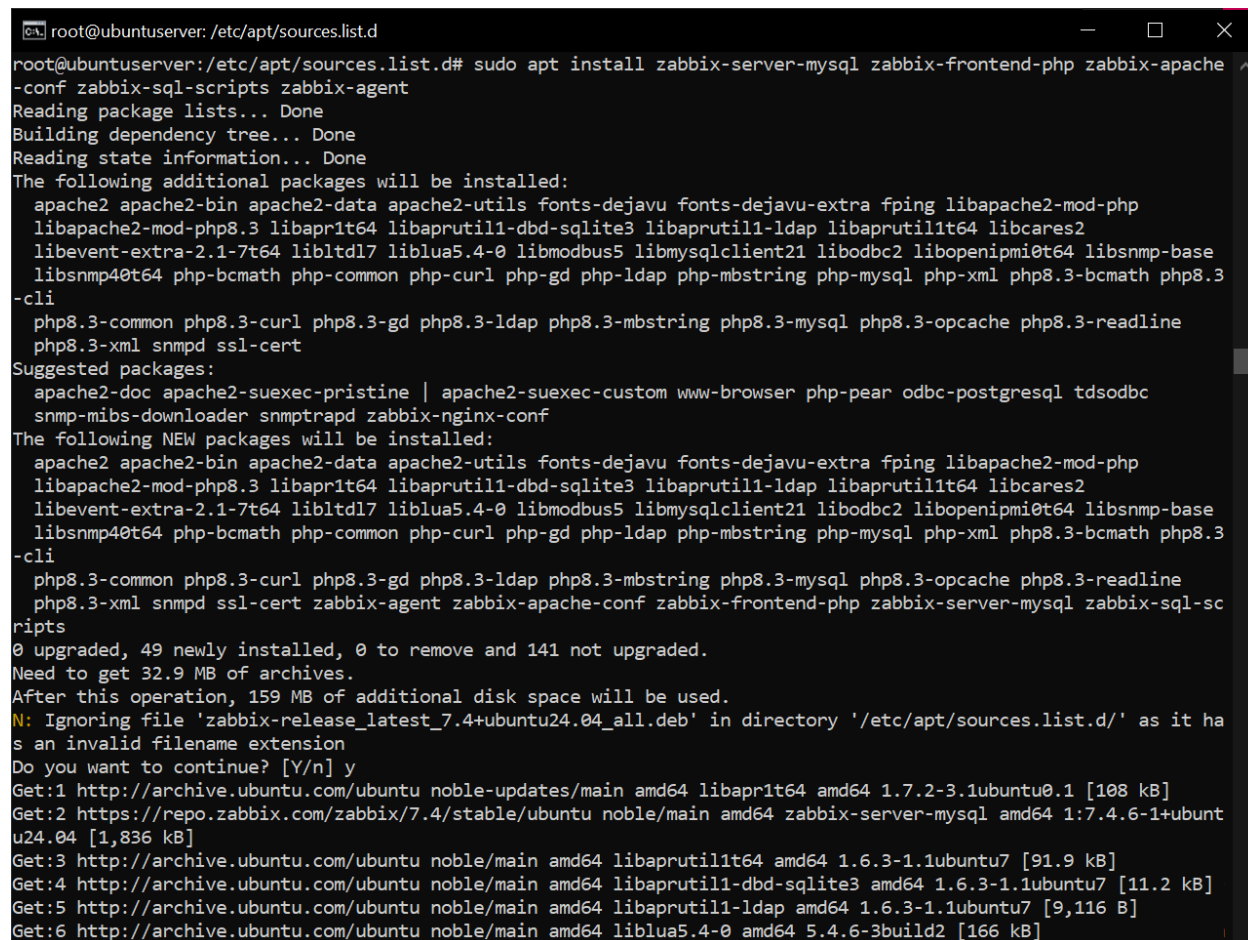
Part 2 – Installation of Zabbix Server

Step 8

Next step would be to Install Zabbix server, frontend, agent to the Ubuntu server you installed.

```
$ sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent -y
```

Screenshot 1



```
root@ubuntuserver: /etc/apt/sources.list.d
root@ubuntuserver:/etc/apt/sources.list.d# sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache
-conf zabbix-sql-scripts zabbix-agent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils fonts-dejavu fonts-dejavu-extra fping libapache2-mod-php
  libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 libcares2
  libevent-extra-2.1-7t64 libltdl7 liblua5.4-0 libmodbus5 libmysqlclient21 libodbc2 libopenipmi0t64 libsnmp-base
  libsnmp40t64 php-bcmath php-common php-curl php-gd php-ldap php-mbstring php-mysql php-xml php8.3-bcmath php8.3
-cli
  php8.3-common php8.3-curl php8.3-gd php8.3-ldap php8.3-mbstring php8.3-mysql php8.3-openssl php8.3-readline
  php8.3-xml snmpd ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser php-pear odbc-postgresql tdsodbc
  snmp-mibs-downloader snmptrapd zabbix-nginx-conf
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils fonts-dejavu fonts-dejavu-extra fping libapache2-mod-php
  libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 libcares2
  libevent-extra-2.1-7t64 libltdl7 liblua5.4-0 libmodbus5 libmysqlclient21 libodbc2 libopenipmi0t64 libsnmp-base
  libsnmp40t64 php-bcmath php-common php-curl php-gd php-ldap php-mbstring php-mysql php-xml php8.3-bcmath php8.3
-cli
  php8.3-common php8.3-curl php8.3-gd php8.3-ldap php8.3-mbstring php8.3-mysql php8.3-openssl php8.3-readline
  php8.3-xml snmpd ssl-cert zabbix-agent zabbix-apache-conf zabbix-frontend-php zabbix-server-mysql zabbix-sql-sc
ripts
0 upgraded, 49 newly installed, 0 to remove and 141 not upgraded.
Need to get 32.9 MB of archives.
After this operation, 159 MB of additional disk space will be used.
N: Ignoring file 'zabbix-release_latest_7.4+ubuntu24.04_all.deb' in directory '/etc/apt/sources.list.d/' as it ha
s an invalid filename extension
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 https://repo.zabbix.com/zabbix/7.4/stable/ubuntu noble/main amd64 zabbix-server-mysql amd64 1:7.4.6-1+ubunt
u24.04 [1,836 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9,116 B]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
```

Part 3 - Zabbix Database Installation

Step 7

On Zabbix server host import initial schema and data. You will be prompted to enter your newly created password from step 4 above. In my own case, I used zabbix as my password.

First change your directory using the command

```
# cd /usr/share/zabbix-sql-scripts/mysql/
```

Then run the command import initial schema and data

```
# zcat server.sql.gz | mysql Zabbix
```

Screenshot 2

```
root@ubuntuuser:/usr/share# sudo zcat /usr/share/zabbix/sql-scripts/mysql/server.sql.gz | mysql --default-character-set=utf8mb4 -uzabbix -p zabbix
Enter password:
root@ubuntuuser:/usr/share#
```

Step 9

Save the configuration file using ctrl o on your keyboard and use ctrl x to exit, then start the Zabbix server and agent processes and check the status using the command below. The status should show **active (running)**.

Command to restart the Zabbix server, Zabbix agent and apache2

```
# systemctl restart zabbix-server zabbix-agent apache2
```

Command to enable the Zabbix server, Zabbix agent and the apache2

```
# systemctl enable zabbix-server zabbix-agent apache2
```

Screenshot 3

```
root@ubuntuuser:/etc/zabbix# root@ubuntuuser:/etc/zabbix# sudo systemctl restart zabbix-server zabbix-agent apache2
root@ubuntuuser:/etc/zabbix# systemctl enable zabbix-server zabbix-agent apache2
Synchronizing state of zabbix-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable zabbix-server
Synchronizing state of zabbix-agent.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable zabbix-agent
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
Created symlink /etc/systemd/system/multi-user.target.wants/zabbix-server.service → /usr/lib/systemd/system/zabbix-server.service.
root@ubuntuuser:/etc/zabbix#
```

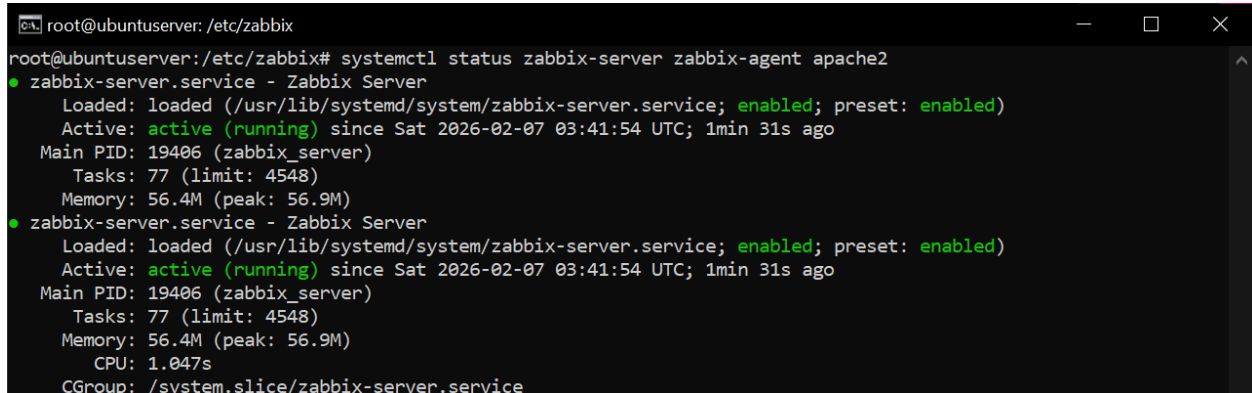
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Command to check the status of your zabbix server, zabbix agent and apache2. Note use the down button on your keyboard to see the other agents running

```
# systemctl status zabbix-server zabbix-agent apache2
```

Zabbix server running

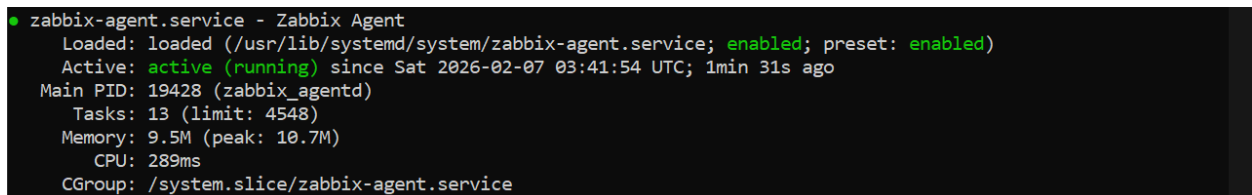
Screenshot 4



```
root@ubuntu: /etc/zabbix
root@ubuntu: /etc/zabbix# systemctl status zabbix-server zabbix-agent apache2
● zabbix-server.service - Zabbix Server
   Loaded: loaded (/usr/lib/systemd/system/zabbix-server.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-02-07 03:41:54 UTC; 1min 31s ago
     Main PID: 19406 (zabbix_server)
        Tasks: 77 (limit: 4548)
      Memory: 56.4M (peak: 56.9M)
● zabbix-agent.service - Zabbix Agent
   Loaded: loaded (/usr/lib/systemd/system/zabbix-agent.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-02-07 03:41:54 UTC; 1min 31s ago
     Main PID: 19428 (zabbix_agentd)
        Tasks: 13 (limit: 4548)
      Memory: 9.5M (peak: 10.7M)
         CPU: 289ms
    CGroup: /system.slice/zabbix-agent.service
```

Zabbix agent running

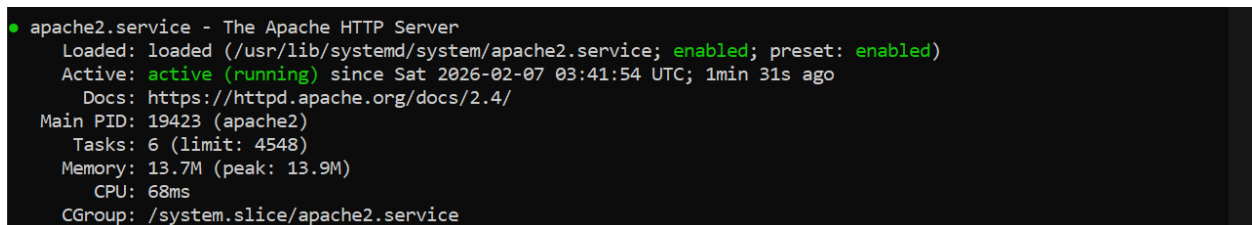
Screenshot 5



```
● zabbix-agent.service - Zabbix Agent
   Loaded: loaded (/usr/lib/systemd/system/zabbix-agent.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-02-07 03:41:54 UTC; 1min 31s ago
     Main PID: 19428 (zabbix_agentd)
        Tasks: 13 (limit: 4548)
      Memory: 9.5M (peak: 10.7M)
         CPU: 289ms
    CGroup: /system.slice/zabbix-agent.service
```

Apache2 service running

Screenshot 6



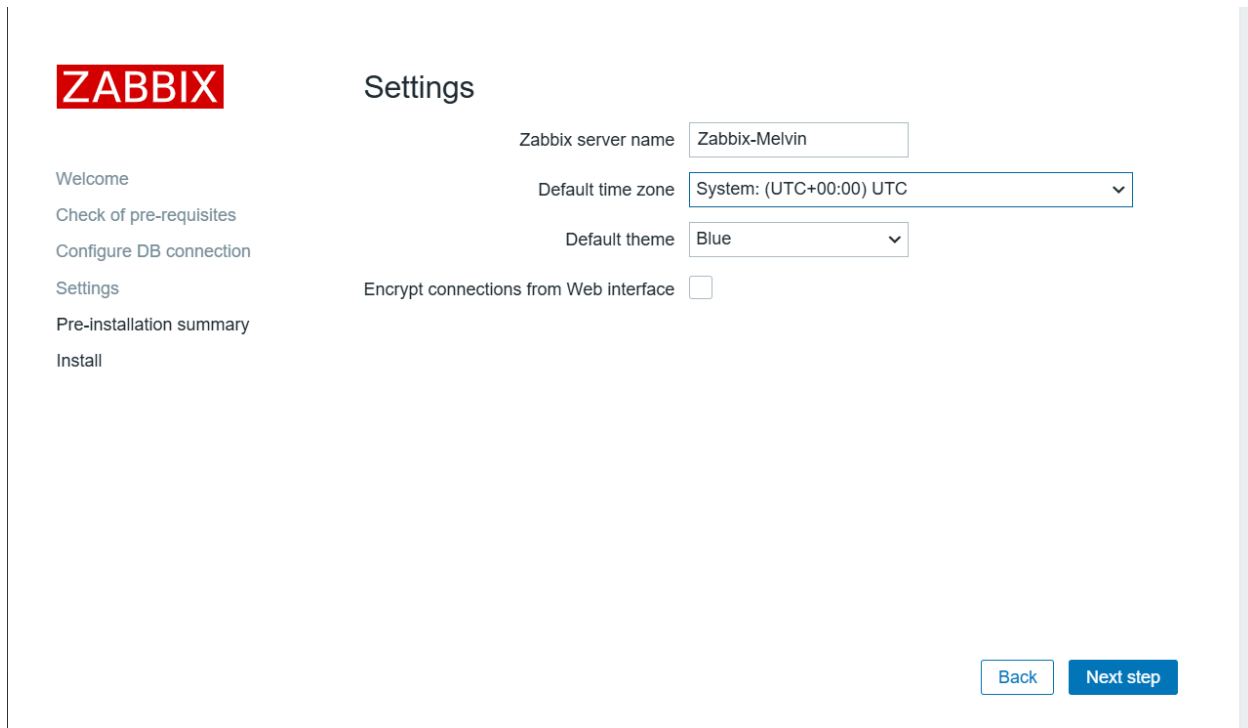
```
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-02-07 03:41:54 UTC; 1min 31s ago
     Docs: https://httpd.apache.org/docs/2.4/
     Main PID: 19423 (apache2)
        Tasks: 6 (limit: 4548)
      Memory: 13.7M (peak: 13.9M)
         CPU: 68ms
    CGroup: /system.slice/apache2.service
```

Part 4 - Access to the Zabbix Web Interface

Step 5

Make your Zabbix server name your name with this naming convention “Zabbix-Yourname” for the Zabbix server name. In my case I used Zabbix-Fuad. Then lick next

Screenshot 7



The screenshot shows the Zabbix web interface during installation. On the left is a sidebar with the ZABBIX logo and a list of steps: Welcome, Check of pre-requisites, Configure DB connection, Settings (highlighted), Pre-installation summary, and Install. The main area is titled 'Settings' and contains three configuration options: 'Zabbix server name' with a text input field containing 'Zabbix-Melvin', 'Default time zone' with a dropdown menu showing 'System: (UTC+00:00) UTC', and 'Default theme' with a dropdown menu showing 'Blue'. Below these is a checkbox for 'Encrypt connections from Web interface' which is currently unchecked. At the bottom right are two buttons: 'Back' and 'Next step'.

ZABBIX

Welcome
Check of pre-requisites
Configure DB connection
Settings
Pre-installation summary
Install

Settings

Zabbix server name

Default time zone

Default theme

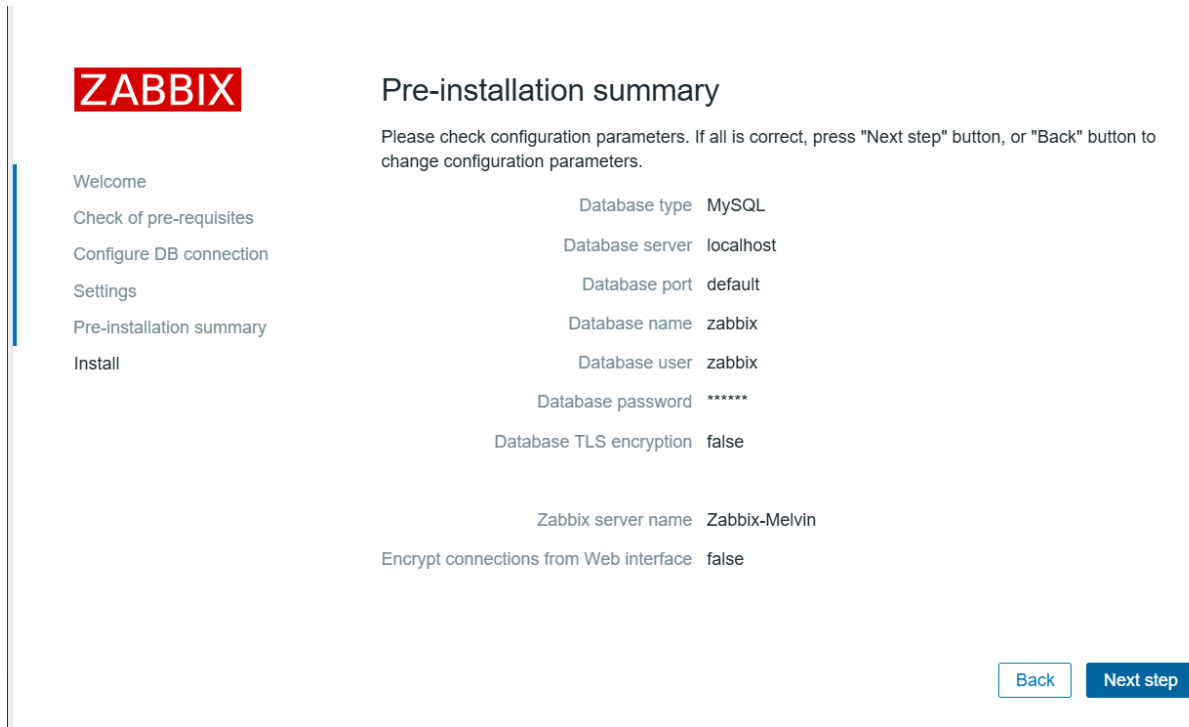
Encrypt connections from Web interface ☐

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Step 6

Check the pre-installation summary to make sure there are no errors.

Screenshot 8



The screenshot shows the ZABBIX pre-installation summary screen. On the left is a vertical navigation menu with the ZABBIX logo at the top. The menu items are: Welcome, Check of pre-requisites, Configure DB connection, Settings, Pre-installation summary (which is highlighted with a blue bar), and Install. The main content area is titled 'Pre-installation summary' and contains a paragraph: 'Please check configuration parameters. If all is correct, press "Next step" button, or "Back" button to change configuration parameters.' Below this is a list of configuration parameters: Database type: MySQL, Database server: localhost, Database port: default, Database name: zabbix, Database user: zabbix, Database password: ***** (masked), Database TLS encryption: false, Zabbix server name: Zabbix-Melvin, and Encrypt connections from Web interface: false. At the bottom right are two buttons: 'Back' and 'Next step'.

ZABBIX

Pre-installation summary

Please check configuration parameters. If all is correct, press "Next step" button, or "Back" button to change configuration parameters.

Database type MySQL

Database server localhost

Database port default

Database name zabbix

Database user zabbix

Database password *****

Database TLS encryption false

Zabbix server name Zabbix-Melvin

Encrypt connections from Web interface false

Back Next step

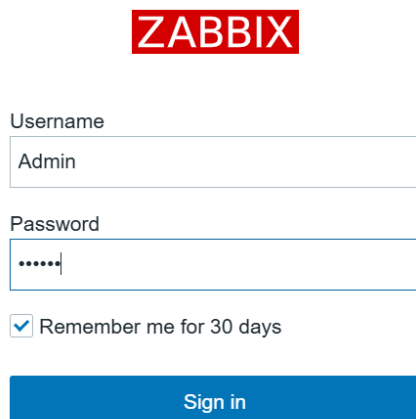
Step 8

You will notice your server's name has changed to the name you added. Pretty cool right!! Input the default zabbix username and password to login.

Username: Admin

Password: zabbix

Screenshot 9



The screenshot shows the ZABBIX login screen. At the top is the ZABBIX logo. Below it are two input fields: 'Username' with the value 'Admin' and 'Password' with masked characters '.....'. Below the password field is a checkbox labeled 'Remember me for 30 days' which is checked. At the bottom is a blue button labeled 'Sign in'.

ZABBIX

Username

Admin

Password

.....

☒ Remember me for 30 days

Sign in

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Screenshot 10

